

## ACID-BASE CHEMISTRY

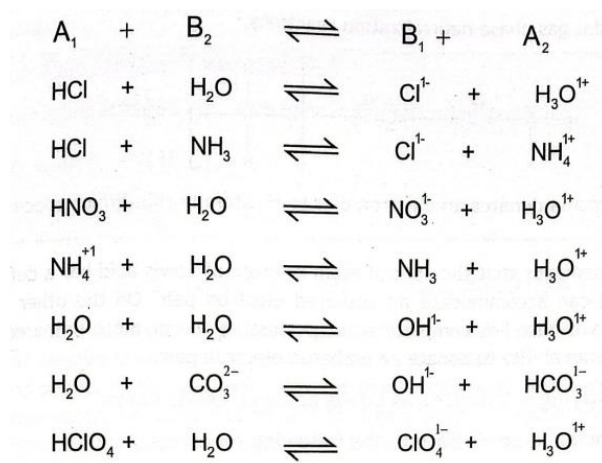
### Conjugate Acid:

A conjugate acid is a species formed as a result of the acceptance of a proton by a base.

### Conjugate base:

A conjugate base is the species that remains after donating a proton from an acid.

- ✓ The weaker acid has stronger conjugate base.
- ✓ Stronger acid has weaker conjugate base.



### Lewis Concept of Acid and Base:

Certain substances like  $SO_2$ ,  $CO_2$ ,  $CaO$ ,  $BF_3$  etc. behave as acids or bases although they do not have ability to donate or accept protons.

Nature of such substances cannot be explained by Arrhenius theory or the Bronsted-Lowry theory.

In 1923, G.N Lewis proposed an acid base theory that focuses on reaction.

### Lewis Acid:

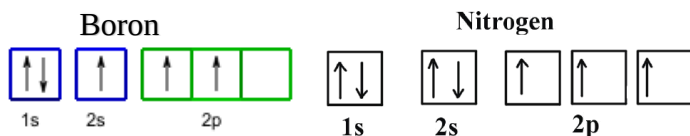
A Lewis acid is substance that can accept a pair of electrons to form a coordinate covalent bond.

### Lewis base:

A Lewis base is a substance that can donate a pair of electrons to form a coordinate covalent bond.

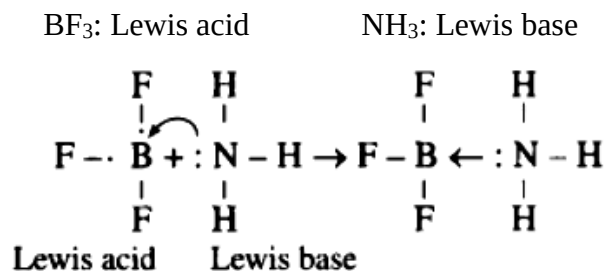
### Bonding between $NH_3$ and $BF_3$ :

The electronic configuration of N and B is following



After,  $BF_3$  formation B has still incomplete octet.

This octet is completed by the electron pair donated by Ammonia and coordinate covalent bond is formed between them.

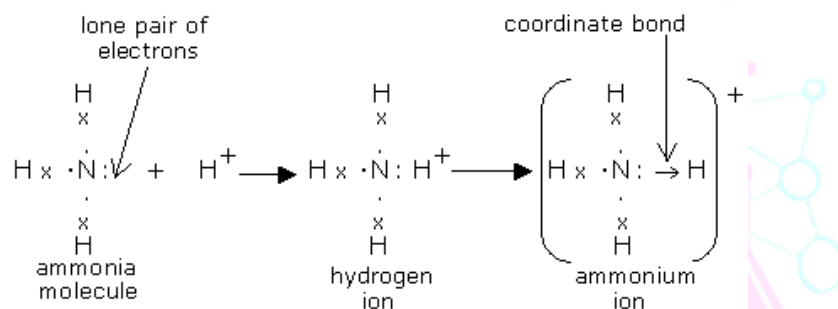


### Formation of Ammonium ion (NH<sub>4</sub><sup>+</sup>):

During the formation of hydronium ion NH<sub>3</sub> containing lone pair of electrons donate a pair of electron to the electron deficient hydrogen atom and form the dative covalent bond.

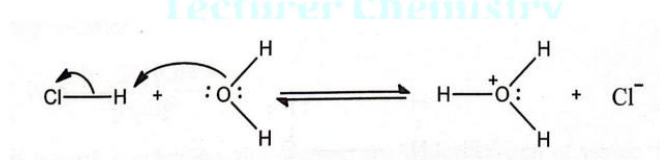
NH<sub>3</sub>: Lewis base

H<sup>+</sup>: Lewis acid



### Bonding between water and hydrogen ion:

Water has lone pair of electrons, it donate its lone of electron to HCl so water acts as Lewis base and HCl Lewis acid.



In a Lewis acid-base reaction a coordinate covalent bond is formed between the acid and the base. Consider the following reaction.

### Titration

*"It is defined as a method to find the volume of the standard solution required to react completely with known volume of another solution under analysis".*

- ✓ Acid-base titrations are conducted using burettes and volumetric pipettes.
- ✓ Generally acid solution is placed in the burette.

- ✓ A fixed volume of base is placed into a conical flask along with a few drops of the suitable acid-base indicator.
- ✓ Acid from the burette is added to the base until the indicator changes colour.
- ✓ This change of colour indicates the end point of titration. In the neutralization of a strong acid with a strong

$$M_1V_1/n_1 = M_2V_2/n_2$$

Where

$M_1$  = Molarity of the base

$V_1$  = Volume of the base taken in flask

$M_2$  = Molarity of the acid

$V_2$  = Volume of acid used from the burette

$n_1$  = No. of moles of base

$n_2$  = No. of moles of acid

### **Selection of Indicators**

#### **1. Strong acid with a Strong base**

When titrating a **strong acid with a strong base**, the pH changes rapidly from 3 to 11. The indicator phenolphthalein changes colour from pH 8 to 10. Therefore, phenolphthalein is the best choice for this titration.

#### **2. Weak acid with a strong base**

When titrating a **weak acid with a strong base**, the pH value changes slightly at the end point. The pH value changes from 6.5 to 10. Since the application range of phenolphthalein is pH 8.3-10, phenolphthalein is the best choice.

#### **3. Strong acid with a weak base**

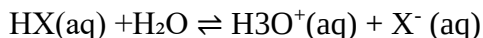
When titrating a **strong acid with a weak base** such as HCl vs  $\text{Na}_2\text{CO}_3$ . At the equivalent point, the pH value changes from 3.5 to 7.5. Methyl orange is the best indicator for this titration.

#### **4. Weak acid and a weak base**

When titrating a **weak acid and a weak base**, since there is no sharp change in the end point, neither indicator works satisfactorily.

**Strength of Acid and Base****1. Strong or weak acid**

The strength of an acid is generally expressed in terms of the acid ionization constant,  $K_a$ . Consider the case of ionization of a general acid  $HX$  in water. In this aqueous solution, the established equilibrium may be represented as follows:



The equilibrium constant  $K$  for this ionization

$$K = \frac{[H_3O^+][X^-]}{[HX][H_2O]}$$

$$K \times [H_2O] = \frac{[H_3O^+][X^-]}{[HX]} \quad (\text{Because water is in excess})$$

$$K_a = \frac{[H_3O^+][X^-]}{[HX]}$$

$K_a$  = Acid dissociation constant

- ✓ Value of  $K_a$  depends upon the temperature.
- ✓ Greater is the value of  $K_a$  stronger is the acid.

**Relationship between  $K_a$  and  $pK_a$** 

$$pK_a = -\log(k_a)$$

Smaller the value of  $pK_a$ , stronger will be the acid.

Table 9.2: Ionisation constants and  $pK_a$  of Acids

| Name of Acid      | Formula      | $K_a$                 | $pK_a$ |
|-------------------|--------------|-----------------------|--------|
| Perchloric acid   | $HClO_4$     | $1.0 \times 10^{10}$  | -10.0  |
| Hydroiodic acid   | $HI$         | $1.0 \times 10^{10}$  | -10.0  |
| Hydrobromic acid  | $HBr$        | $1.0 \times 10^9$     | -9.0   |
| Hydrochloric acid | $HCl$        | $1.0 \times 10^6$     | -6.0   |
| Sulphuric acid    | $H_2SO_4$    | $1.0 \times 10^3$     | -3.0   |
| Hydrofluoric acid | $HF$         | $7.2 \times 10^{-4}$  | +3.1   |
| Formic acid       | $HCOOH$      | $1.8 \times 10^{-4}$  | +3.75  |
| Benzoic acid      | $C_6H_5COOH$ | $6.3 \times 10^{-5}$  | +4.2   |
| Acetic acid       | $CH_3COOH$   | $1.8 \times 10^{-5}$  | +4.7   |
| Phenol            | $C_6H_5OH$   | $1.3 \times 10^{-10}$ | +8.9   |
| Water             | $H_2O$       | $1.8 \times 10^{-16}$ | +15.7  |

**2. Strong or weak Base:**

The ability of a base to accept a proton from an acid, usually water, is termed as strong base.

For a base  $B$ , an equilibrium reaction with water can be represented by the following equation:



The equilibrium constant  $K$  is referred to as base ionization constant and can be derived in the same way as  $K_a$  for acids. Thus:

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

- ✓  $K_b$  value will be large if degree of ionization of the base  $B$  is high i.e. if the base  $B$  is strong.
- ✓ The Value of  $K_b$  will be small for a weak base  $B$ .
- ✓ Again, for convenience, a parameter  $pK_b$ , has been devised to express  $K$  value in convenient numbers.
- ✓ Thus,  $pK_b$  is defined as the negative logarithm of  $K_b$ .

$$pK_b = -\log K_b$$

Table 9.3:  $K_b$  and  $pK_b$  Values of Some Common Bases

| Name of Base | Formula        | $K_b$                 | $pK_b$ |
|--------------|----------------|-----------------------|--------|
| Diethylamine | $(C_2H_5)_2NH$ | $9.6 \times 10^{-4}$  | 3.02   |
| Ethylamine   | $C_2H_5NH_2$   | $5.6 \times 10^{-4}$  | 3.25   |
| Methylamine  | $CH_3NH_2$     | $4.5 \times 10^{-4}$  | 3.34   |
| Ammonia      | $NH_3$         | $1.7 \times 10^{-5}$  | 4.76   |
| Pyridine     | $C_5H_5N$      | $5.6 \times 10^{-9}$  | 8.25   |
| Aniline      | $C_6H_5NH_2$   | $4.3 \times 10^{-10}$ | 9.37   |

### Buffer solution and their Applications

#### Buffer solution:

*"A solution which resists the change in pH is called buffer solution."*

#### Types of Buffer solution

##### Acidic Buffer:

It is formed by mixing a weak acid and salt of it (with strong base). Such solution give acidic buffer with pH less than 7.

**Example:**  $CH_3COOH + CH_3COONa$

##### Basic Buffer:

It is formed by mixing a weak base and salt of it (with strong acid). Such solution give basic buffer with pH more than 7.

**Example:**  $NH_4OH + NH_4Cl$

#### Calculation of pH of Buffer solution

pH of a buffer can be calculated by using **Henderson's- Hasselbalch equation**.

##### For Acid Buffer

$$pH = pK_a + \log \frac{[Salt]}{[Acid]}$$

**For Basic Buffer**

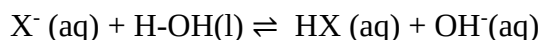
$$\text{pH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]}$$

**Salt Hydrolysis**

*"The chemical reaction of cations and anions of salts with water is called hydrolysis. "*

*"In hydrolysis H - OH bond is broken whereas in hydration water molecule adds up to a substance without bond breakage."*

When a salt (MX) is dissolved in water, it splits up into its M and X ions.

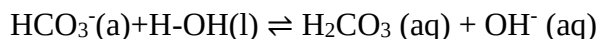
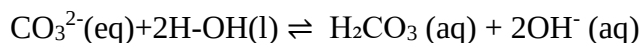
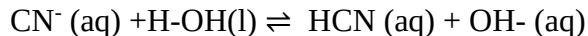
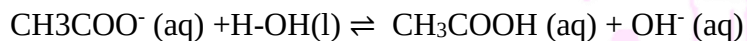
**Reaction of M and X ions with water**

Since  $\text{H}^+$  and  $\text{OH}^-$  ions are produced in these reactions, the solution of the salt may be acidic or basic.

In salts anions are derived from acids and cations from bases.

**Basic Solution:**

The anions of weak acids are strong conjugate bases with water producing basic solutions.

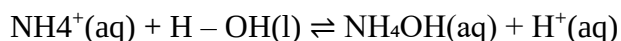
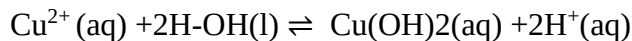


Anions like  $\text{Cl}^-$ ,  $\text{NO}_3^-$ ,  $\text{SO}_4^{2-}$ , are so weak conjugate bases that, they do not react with water.

**Acidic Solution**

Cations of weak bases are strong conjugate acids react with water producing acidic solutions.

For example:  $\text{Cu}^{2+}$ ,  $\text{Al}^{3+}$ ,  $\text{NH}_4^+$ , etc.



Cations like  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ , etc. are so weak conjugate acids that they do not react with water.

**Types of Salts:**

There are four types of salts on the basis of their reactivity with water:

1. Salts of **strong acids and strong bases** do not hydrolyze ( $\text{pH} = 7$ ).

Examples:  $\text{NaCl}$ ,  $\text{Na}_2\text{SO}_4$ ,  $\text{KNO}_3$  etc.

2. Salts of **weak acids and strong bases** hydrolyze producing basic solutions ( $\text{pH} > 7$ ).

Examples:  $\text{CH}_3\text{COONa}$ ,  $\text{NaCN}$ ,  $\text{Na}_2\text{S}$  etc.

3. Salts of **strong acids and weak bases** hydrolyze producing acidic solutions ( $\text{pH} < 7$ ).

Examples:  $\text{CuSO}_4$ ,  $\text{NH}_4\text{Cl}$ ,  $\text{NH}_4\text{NO}_3$ , etc.

4. Salts of **weak acids and weak bases** hydrolyse, but the resulting solution is either neutral, acidic or basic. This depends upon the relative values of  $K_a$  and  $K_b$  of cations and anions of the salt.

Table 9.5

| Salt Type |        | Common Example                                    | Ions which impart Hydrolysis | Solution pH (Nature)    |                                                   |                  |                                           |
|-----------|--------|---------------------------------------------------|------------------------------|-------------------------|---------------------------------------------------|------------------|-------------------------------------------|
| Acid      | Base   |                                                   |                              |                         |                                                   |                  |                                           |
| Strong    | Strong | $\text{NaCl}$ , $\text{KBr}$                      | None                         | $\approx 7.0$ (Neutral) |                                                   |                  |                                           |
| Strong    | Weak   | $\text{NH}_4\text{NO}_3$ , $\text{NH}_4\text{Cl}$ | Cations                      | $< 7.0$ (Acidic)        |                                                   |                  |                                           |
| Weak      | Strong | $\text{NaCN}$ , $\text{K}_2\text{CO}_3$           | Anions                       | $> 7.0$ (Basic)         |                                                   |                  |                                           |
| Weak      |        |                                                   |                              | Weak                    | $\text{NH}_4\text{CN}$ , $\text{NH}_4\text{NO}_2$ | Anions & Cations | May be equal, smaller or greater than 7.0 |

### Solubility Product

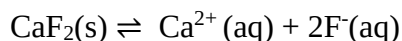
It is defined as

*“The product of the equilibrium concentrations of ions, each raised to a power which is the coefficient of the ion in the balance chemical equation.”*

When an excess of a sparingly soluble ionic compound is mixed with water. Some of it dissolves and the rest of the compound settles to the bottom.

A dynamic equilibrium is reached between an insoluble solid compound and its ions in a saturated solution.

**For example**, when it is mixed with water. The following equilibrium occurs.



$K_c$  for this equilibrium can be written as

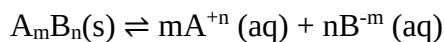
$$K = \frac{[\text{Ca}^{2+}][\text{F}^{-}]^2}{[\text{CaF}_2]}$$

As  $\text{CaF}_2$  is slightly soluble so its concentration remains constant.

$$K[\text{CaF}_2] = [\text{Ca}^{2+}][\text{F}^{-}]^2$$

$$K_{sp} = [\text{Ca}^{2+}][\text{F}^{-}]^2$$

**In general**,  $K_{sp}$  expression of any slightly soluble ionic compound  $A_mB_n$ , can be written as



$$K_{sp} = [\text{A}^{+n}]^m [\text{B}^{-m}]^n$$

### Precipitation Reactions

*“An aqueous reaction that takes place when two or more solution are mixed together, yielding a solid insoluble substance is called precipitation reaction.”*

It is obtained by substituting initial concentrations instead of equilibrium concentrations in the expression for  $K_{sp}$ .

For example, ion product expression for solid  $\text{CaF}_2$  is given by

$$Q' = [\text{Initial conc. of } \text{Ca}^{2+}] [\text{Initial conc. of } \text{F}^{1-}]^2$$

If we add a solution containing  $\text{Ca}^{2+}$  ions to a solution containing  $\text{F}^-$  ions, precipitate may or may not form.

To predict whether a precipitation will occur, we compare  $Q'$  and  $K_{sp}$ . There are two possibilities.

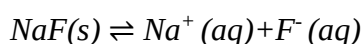
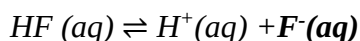
- If  $Q' > K_{sp}$ , precipitation occurs and will continue until the concentration satisfy  $K_{sp}$ .
- If  $Q' < K_{sp}$ , precipitation does not occur.

### Common Ion effect

*“The phenomenon in which the degree of ionization or solubility of an electrolyte is reduced by the addition of a highly soluble electrolyte containing a common ion is called the common ion effect.”*

#### Explanation:

For example, in a solution of weak acid, hydrofluoric acid  $K_a = 7.2 \times 10^{-4}$ , its salt sodium fluoride produces the common ion.



A strong electrolyte breaks up completely into its ions.  $\text{HF}$  is a weak electrolyte, it dissociates slightly.  $\text{NaF}$  being strong electrolyte, completely dissociates in its ions. The common ion  $\text{F}^-$  produced by  $\text{NaF}$  disturbs this balance. This increases the concentration of  $\text{F}^-$  ions.

According to **Le Chatelier's principle**, the equilibrium shifts to the left to use up some of the  $\text{F}^-$  ions. This reduces  $\text{HF}$  dissociation. Thus, the dissociation of  $\text{HF}$  is reduced in the presence of dissolved  $\text{NaF}$ . This means that the  $\text{HF}$  concentration increases as a result of the equilibrium change.

Similarly, when a highly soluble salt is added to a saturated solution of a less soluble salt containing a common ion. The degree of dissociation of the less soluble salt decreases. Therefore, it reduces its solubility. The term common ion effect is used to describe the behavior in which two different compounds produce the same ion.

**For More Information**

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